KAI DEVITO

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EDUCATION

University of Georgia **Graduation: May 2027**

B.S. in Computer Systems Engineering / Electrical Engineering, Double Dawgs Program; Minor in Computer Science **GPA: 3.86/4.0**

Honors: Morehead Honors College; Tau Beta Pi Engineering Honor Society; Presidential Scholar

WORK EXPERIENCE & RESEARCH

MathWorks Research & Design Team, *University of Georgia* Oct 2026 – Present

- Direct the human following robot project's computer vision architecture supervised by a pre-trained deep learning algorithm
- Interview project stakeholders including faculty, fabrication staff, and engineering students about the project's community impact

USG Student Grader, *University of Georgia* Aug 2025 – Dec 2025

- Graded students' project submissions done using R compiled in Jupyter Notebook for Quality Engineering, a senior-level mechanical engineering elective course, and Advanced Topics in Engineering, a graduate computer/cognitive science research course
- Communicated students with extenuating circumstances, announced project rubrics and expectations, and alerted point deductions for calculation errors, insufficient work, or failures to submit assignments on time

Smart Manufacturing Group, *University of Georgia* Jul 2025 – Nov 2025

- Operated a deep learning quality control system comprising of UR5e and DOBOT Magician Lite robotics arms to detect 3D-printed component defects and have a Gemini AI language model generate the automated inspection report
- Prepared individual research conceptualizing a voice control system for simulating and manipulating Cartesian coordinates built using Python modules for qualification to the team

STUDENT PROJECTS

UGA Computer Architecture Group, *University of Georgia* Aug 2026 – Present

- Improve a preexisting single-cycle computer with RISC ISA by writing new 32-bit Assembly instructions for the processor to be more resourceful with memory and put less stress on the Python assembler and Rust emulator
- Design new instructions for supporting a stack data structure and thus streamlining computer memory for recursive arithmetic and branching operations for 16 registers

Sensors Interactive Robotic Pet (Video Link), *University of Georgia* May 2026 – Aug 2026

- Developed an embedded sensors system using an ESP32 Developer's Kit that changes emotional data from a raw LDR, two prepackaged IC sensors, and a homemade touch sensor to simulate a robotic pet toy
- Built an IoT data pipeline that fetched ESP32 data as JSON objects over WLAN with a JavaScript file implementing a local host server that posts data to MongoDB from Node.js' HTTP methods to have a user-friendly interface of live data

UGA Design Methodology Team, *University of Georgia* Jan 2026 – May 2026

- Designed and constructed an embedded electrical measurement toolbench in Python using a Raspberry Pi 4 and SAR ADC for ohmmeter, voltmeter, voltage reference, waveform generation, and frequency measurement
- Developed Python firmware to integrate electronic components through I2C and SPI, including an LCD display and rotary encoders
- Coordinated meetings to complete electrical testing, hardware debugging and update the technical documentation and user manual

SKILLS

Languages & HDL: C, C++, ARM64 Assembly, Icarus Verilog, Verilog HDL, Rust, Python, JavaScript, TypeScript, MATLAB, R

CAD & 3D Modeling: Tinkercad, Fusion 360, BambuStudio, BrickLink Studio (P1S 3D Printer), Blender

Embedded Systems: FPGA Basys 3, Raspberry Pi 4, ESP32, WLAN, SPI, I2C, Linux kernel

Hardware & Lab: KiCad, PCB Design, MultiSim, WaveForms SDK, Oscilloscopes, Breadboarding, Soldering, Git